# Compactness for the sequent calculus

The Open Logic Project contributors

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*Editorial scope.* Statement and complete author proof are included verbatim as the span lines around ollabel prop:proves-compact of the pinned original source; no text is rewritten or paraphrased.

*Allowed standard prerequisites (editorial).*

- LK derivations of sequents Gamma sequent A
- the entailment relation Gamma proves A read off derivability of the sequent
- inconsistency as derivability of the empty sequent

**Proposition 0.1** (Compactness)**.** *1. If* $\Gamma \vdash !A$ *then there is a finite subset* $\Gamma_0 \subseteq \Gamma$ *such that* $\Gamma_0 \vdash !A$*.*

*2. If every finite subset of* $\Gamma$ *is consistent, then* $\Gamma$ *is consistent.*

*Proof.* 1. If $\Gamma \vdash !A$, then there is a finite subset $\Gamma_0 \subseteq \Gamma$ such that the sequent $\Gamma_0 \Rightarrow !A$ has !!aderivation. Consequently, $\Gamma_0 \vdash !A$.

2. If $\Gamma$ is inconsistent, there is a finite subset $\Gamma_0 \subseteq \Gamma$ such that L$LK$ !!derives $\Gamma_0 \Rightarrow$ . But then $\Gamma_0$ is a finite subset of $\Gamma$ that is inconsistent.

□

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